RMIT UNIVERSITY

Test-Driven Retrieval-Augmented Generation

Woolworths FY25 Report Q&A

WIL Project: Written Report

Group 59

25 October 2026

Decision focus

Can a small, source-grounded financial-report assistant help investors and analysts locate and verify reported figures?

Evidence basis

A reproducible 40-question diagnostic benchmark, measured keyword/dense/hybrid retrieval comparisons, controlled ablation and source-checked reference answers.

Project repository

`https://github.com/s4097104/Data-Science-Assignment`

Project board

Group 59 Trello board

Executive summary

Investment analysts must get financial answers that have the right year, unit, accounting basis, and source. Prototype by Group 59 utilizes the search of evidence combined with locally deployed language model for this purpose. This report enhances the evaluation through performing and reproducing comparison between keyword, dense and hybrid retrieval on the provided Five Year Summary of Woolworths FY25.

The new diagnostic benchmark consists of 40 questions: 12 directly stated facts, 8 paraphrases, 8 comparisons, 4 ambiguous questions and 8 unsupported questions. The metrics for retrieval take into account only 28 answerable questions. Having the same 10 source passages and passage-level relevance labels, BM25 retrieves relevant evidence in its top five for 27/28 questions, dense retrieval for 28/28 and hybrid retrieval for 27/28. NDCG@5 metric equals 0.760, 0.931 and 0.871 correspondingly. Dense retrieves relevant evidence in its first position for 24/28 questions while BM25 does so for 17/28.

BM25 ablation with controlled settings reveals that eliminating passage-length normalization decreases NDCG@5 from 0.760 to 0.668. An experiment with unsupported questions confirms that all eight unsupported questions still receive positive keyword scores, proving why a naive zero-score policy cannot be applied. Token-length analysis shows that out of ten passages, nine surpass the embedding model limit of 512 tokens.

The recommended approach is to use dense retrieval as a pilot baseline, keep human verification and prioritize table-aware passages and generated answer validation. The question set used for evaluation consists of 40 questions written with awareness of the source and correlated paraphrases; labels have been generated with help of AI and need further human verification. There is no data regarding local language model answers and user timings for the evaluation. Historical team-claimed large corpus scores have been kept only for contextual information.

1 Problem definition and achieved aims

1.1 Business problem and stakeholders

Stakeholders would include retail investors analyzing company disclosures and financial analysts creating comparable financial summaries. Practically speaking, their purpose is to reach a definitive answer, not necessarily produce smooth language. In the case of “profit”, it can either mean profit after tax or profit before adjustments. Assuming that they are the same will change the result. Also, net debt will include leases while another debt figure in the same table will exclude them [1].

Benefits include saving time finding evidence, increased visibility of financial definitions, and verifiability via page citations. The prototype is a tool for document navigation and explanations. This does not make any judgments on whether the investment is suitable or what future performance will be like. There is no connection with Woolworths at all.

1.2 Research questions

1. **RQ1: Evidence access.** How effectively does the baseline retrieve the evidence needed for direct, comparison and explanatory financial questions?
2. **RQ2: Retrieval design.** Does combining keyword and semantic retrieval improve evidence ranking compared with either method alone?
3. **RQ3: Answer trustworthiness.** Can the complete system provide numerically correct, supported answers with valid source references and abstain when evidence is insufficient?

4. **RQ4: Stakeholder value.** Does the tool reduce time to a verified answer without increasing factual errors relative to manual PDF search?

RQ1 and RQ2 have been tested using the new controlled retrieval experiment. For RQ3, there is an answer key with sources and behavioral tests available for grading the generated answers but no measured accuracy of the LLM. RQ4 is still untested.

1.3 Scope and relationship to Milestone 1

Question-posing milestone 1 for Woolworths' FY2025 Annual Report and the initial experiments of Walert search have been outlined in Milestone [2]. The configuration of the financial summary prototype restricts the range to the small source. The draft of 24 October mentions a relevant experiment of the annual report closer to the original purpose [6]. They should be considered as two different configurations, and the given sources do not indicate their specific order of realization.

The configuration of the summary involves the issues of revenues, profits, debts, cash flow, shareholder data, and some ratios on the basis of the nine-page summary. The separate larger experiment utilizes the 89-page source, but its particular PDF format and page numbers are still unclear. Neither of the two configurations does not imply the access to any live prices, future financial periods, or facts that are not covered by its corpus.

Smaller scope allows for feasible manual verification, whereas a larger scope allows for testing of retrieval under difficult conditions. This is a practical interpretation of the relevance of the two scopes, not an assertion that a past decision by some team was to go through that sequence. All financial examples in this report are based on the given summary.

2 Methodology

2.1 Data and provenance

The source PDF is a publicly released document containing financial information for the company [1]. Provided `chunks.csv` consists of ten uniquely identified excerpts covering all nine pages; the second page is divided into two parts - a financial table and explanatory notes. The lengths of chunks are between 153 and 939 words. Each record is preserved with chunk id, source name, page number and text. Provided clean financial data CSV is not used by the RAG pipeline in reading step: reading works with `chunks.csv`.

This is an important point since the existence of a structured file does not mean that the model utilizes a valid table look-up. The extraction process can mix footnotes in superscripts with years and split numbers: the extracted New Zealand Food row contains "7, 24 0" while the original PDF shows 7,240. The analysis presented in the work then utilizes the original PDF as a numerical reference point and treats the extracted text as a derived one. No ingestion script or extraction history were provided; thus, the precise reconstruction of the first extraction is not possible.

The audit package contains SHA-256 hashes for the PDF, chunks, questions, and BM25 implementation in addition to the Python environment. This ensures that results are tied to an exact snapshot of the input. Any future update must maintain the year, unit, accounting principle, table heading, and footnotes link for each passage.

2.2 Evaluation scope and historical configurations

The primary experiment employs the nine-page summary provided, along with ten sections. All other aspects of retrieval can then make use of the same question strings and relevance judgments.

The previous experiment using eighteen questions serves as a diagnostic benchmark; its 94.4% page hit performance is not directly comparable to the current 40 question benchmark, since the set of questions is different.

The 24 October team draft independently describes the testing against an 89-page document containing 494 sections and ten questions [6]. These results are not included in the current set of experiments. They could not be included, since the document, output log, and three screenshots cited were missing from the submitted archive. The combined report uses the smaller corpus for confirmed results, while retaining the larger set as an independent experiment.

2.3 Retrieval and generation design

Retrieval with keywords. The provided BM25 variant lower-cases text and selects only alphanumeric words. It encourages important terms in queries while discounting redundant terms and the passage length. Parameters are set to $k_1 = 1.5$ and $b = 0.75$, and by default produces five passages. This inexpensive and explainable approach is suitable for matching precise financial labels. It does not perform synonym, numeric, or accounting concept disambiguation.

Semantic Retrieval. The script calls `multi-qa-MiniLM-L6-cos-v1`. The model card explains that there are 384-dimensional embedding vectors used for semantic search, where an input limit of 512 words is set [7]. In the implementation, document and question vectors are normalized, and then their dot product is computed, which is the same as cosine similarity if the vectors are unit vectors. The token lengths are calculated before truncation: nine out of ten documents are longer than 512 tokens, ranging from 202 to 1,400 tokens. Truncation may cut financial rows or notes at the end even if the whole document is shown to the user. For the new test, the ONNX model file is downloaded from the publication website and run on CPU. Mean pooling on attention mask with subsequent L2 normalization yields 384-dimensional vectors. There is an input limit of 512 tokens because the model uses sentence-transformer setting.

Hybrid ranking. Reciprocal rank fusion combines the top five BM25 and dense candidates using equal weights:

$$S(c) = \frac{0.5}{60 + r_B(c)} + \frac{0.5}{60 + r_D(c)}. \tag{1}$$

The contribution by an omitted candidate is null to that list. Positions are ranked on a sum of combined rather than incomparable scores. It seems like a reasonable compromise between accurate terminology and semantic similarity; the experiment detailed below aims to check its effectiveness. Since each input list is trimmed to five entries, any missing passage cannot be recovered using fusion.

Answer generation. The pipeline sends the query and fused passages to the local Ollama API, using model tag `llama3.2`, temperature zero and a 180 seconds request timeout. The prompt asks the model to rely on provided context, avoid fictional characters, and include pages and to state when the provided context is inadequate. These requests are precautions but not guarantees. There is no additional numeric validator, citation verifier or retrieval abstention threshold. Temperature zero does not negate the requirement to keep the version and runtime information.

Interface. The Streamlit application has a question input area, examples of questions, an output of answers, evidence expansion, and a separated retrieval view for keywords/semantics. This allows for the inspection. Nevertheless, the slider on the side bar is not passed into the cached pipeline that is defined using five sources. Also, the source code sets the question widget session state right after the widget creation once Clear is pressed; this needs to be checked against the selected Streamlit version. These statements are made based on code inspection, not successful browser demo.

2.4 Benchmark construction and evaluation controls

Walert makes the case of evaluation of evidence retrieval and conversational answers independently [4, 5]. The new benchmark includes this idea. It consists of 40 records where there is a stable question ID, question type, label of answerability, source-checked reference answer, relevant chunk IDs, expected behavior, and topic group. There are twelve factual questions, eight paraphrasing of factual questions, and eight questions on comparisons or calculations. Four ambiguous questions and eight unsupported questions are omitted in retrieval denominators.

Units, dates, and accounting bases are preserved in the answers to references. For example, EBIT in the answer is separate from EBIT before major items; net debt in the answer is separate from debt without leases. Seven arithmetic verifications independently validate changes in EBIT, revenue, debt, cash flow, dividends, EPS, and profit reported. This is an arithmetic check of the answer key, not a scoring of language model responses.

Question definition and retrieval configurations were crafted prior to conducting the new comparison. Nevertheless, the source and the original question set have been analyzed, and the new questions contain factual/paraphrase pairs that overlap. This is an AI-augmented retrospective diagnosis set, and not a blind held-out benchmark or an independent human assessment. All labels are second reviewed; no inter-rater agreement is asserted. The next step is a hold-out that separates topic sets, thus paraphrases of the same fact cannot be on both sides of the hold-out.

All major approaches employ the same sets of passages, questions, binary assessments of passage relevance, and cutoffs of one, three, and five. BM25 uses $k_1 = 1.5$, $b = 0.75$; dense ranking uses the specified MiniLM model; hybrid ranks combine the top five results from each approach using equal weighting of RRF, with a fixed 60 for all queries. Ties are broken based on the original order in the harness. None of the methods are fine-tuned after calculating the new scores.

Two additional lexical controls are also used: the TF-IDF weights for word unigrams and bigrams together with cosine similarity, and RRF for BM25 and TF-IDF. The former control is called lexical fusion; the latter does not use semantic hybrid retrieval. The control modification is in changing the value of the length normalization parameter in BM25 ($b = 0.75 \to b = 0$).

2.5 Metrics and correction of the starter evaluator

For question q, let R_q be all relevant passages in the corpus, L_q^5 the five retrieved passages, and r_q the first relevant rank. Hit@5 records whether any relevant passage is found; Recall@5 records $|R_q \cap L_q^5|/|R_q|$. MRR@5 averages $1/r_q$, or zero if no relevant passage appears. NDCG@5 rewards relevant evidence appearing earlier:

$$\text{DCG@5} = \sum_{i=1}^{5} \frac{rel_i}{\log_2(i+1)}, \qquad \text{NDCG@5} = \frac{\text{DCG@5}}{\sum_{i=1}^{\min(5,|R_q|)} 1/\log_2(i+1)}. \tag{2}$$

The supplied evaluator builds its ideal ranking from only retrieved labels. This can overstate quality when relevant passages are missing. For example, one relevant passage at rank one receives 1.000 under that approach even if two relevant passages exist. The corrected score is $1/(1 + 1/\log_2 3) = 0.613$. A regression check in the audit script verifies this case. This report uses the corrected denominator; it does not silently replace the original application evaluator.

Two team members should independently label reference answers and relevance for reconciliation and subsequent experiments, reconciling discrepancies and saving both the original and reconciled labels. For future rounds of experiments, fixate the development split prior to tuning and produce a separate holdout split with questions that cover paraphrasing, comparison, ambiguity and unsupported queries. Compare BM25, dense and hybrid models with identical inputs, model, prompt and cut-off. Perform paired question-level comparisons and report uncertainties; avoid concluding significance

Table 1: End-to-end evaluation protocol. These measures are specified, not yet measured.

Dimension	Operational measure	Evaluation control
Answer correctness	Correct responses / answerable questions; exact year, value, unit and accounting basis required	Gold answers checked against PDF; derived values checked with deterministic arithmetic
Unanswered rate	Abstentions / all questions; also report false abstentions on answerable questions	Separate unsupported questions from retrieval failures
Abstention quality	Correct refusals / unsupported questions	Add absent-period, live-price and absent-topic questions
Faithfulness	Supported factual claims / all factual claims	Human claim-level review; no credit for plausible but unsupported facts
Attribution	Supported cited claims / cited claims; plus fraction of factual claims with support	Check actual cited passage, not merely a valid page number
Usability and efficiency	Correct task completion; median and 90th-percentile time to verified answer	Counterbalanced manual-search versus RAG study; include checking time

based on a single question for language model inference. Human judged correctness trumps an unvalidated language model judge.

2.6 Tests for unsupported and ambiguous behavior

The eight prompts in the category of unsupported queries refer to facts that lie outside of the data source – live prices, future scores, individual store sales etc. Four prompts fall into the category of ambiguity, namely profit, debt, dividends and EBIT. Reference behaviors for these prompts imply rejection or clarification. They don't have invented gold evidence. A diagnostic test for zero overlap for BM25 and TF-IDF involves rejecting only when the highest retrieval score is zero.

The correctness, faithfulness, citations, and clarification of the generated answers have not been scored since there are no responses from the LLM. The package contains a scoring template and a separate runner for running it on the team's Ollama environment. The results should never be reported as done without any actual generated outputs being scored.

3 Findings

3.1 Controlled comparison on 28 answerable questions

Table 2: New benchmark results. All methods use the same 28 answerable questions and ten passages.

Method	Hit@1	Hit@5	Recall@5	MRR@5	NDCG@5
BM25	17/28	27/28	0.929	0.727	0.760
Dense (ONNX)	24/28	28/28	0.982	0.917	0.931
Hybrid BM25+dense	21/28	27/28	0.964	0.845	0.871

Dense ranks highest on this benchmark, with an NDCG@5 score that beats BM25 by 0.171 and hybrid by 0.060. Compared to BM25, dense outperforms on 13 questions, matches on 14 and underperforms on one; hybrid wins on 12, matches on 14 and underperforms on two. Such figures

make the differential improvement clear. This small, correlated diagnostic set cannot be used to draw a significance argument for the population.

While the perfect Hit@5 for dense (28/28) does not mean perfect evidence recovery (Recall@5 = 0.982 due to missing one alternative relevant passage), nor does it mean 100% answer accuracy. Five out of ten is a generous cut-off point, so performance at top-one and top-three positions becomes relevant. At k = 3, BM25 performs correctly on 21/28 questions, dense on 28/28 and hybrid on 27/28. Therefore, dense delivers useful evidence using fewer context slots.

3.2 Paraphrases and additional baselines

For 12 direct questions, the NDCG@5 of BM25 is 0.824, for eight paraphrases 0.610, and for eight comparison questions 0.813. The poor results on paraphrases confirm that exact terminology can be a barrier. Comparison questions need an evidence interpreter - table retriever cannot verify if calculations have been done correctly.

TF-IDF attains 25/28 and 0.695 at NDCG@5. Lexical fusion attains 27/28 and 0.718. These baselines give no indication that any benefit would arise from combination of lexical approaches and outperform BM25. Neither does the main hybrid perform better than dense approach. Justifiable complexity implies that its gains exceed its costs; in our case, it supports dense as pilot's retrieval baseline.

3.3 Execution of ablation study: passage-length normalisation

The research hypothesis is that passages dealing with finances may be helped by adjustment of long passage penalty. The test changes only b coefficient from 0.75 to zero. While Hit@5 is still 27/28, Hit@1 decreases from 17/28 to 12/28, MRR@5 from 0.727 to 0.601 and NDCG@5 from 0.760 to 0.668. One NDCG result improves, 18 remain tied and nine worsen.

This failed change provides useful insight: five highest coverage would hide the problem of ranking. Keeping the original length correction is the right evidence-based decision. The experiment is the record of a real hypothesize-test-decision process done in the current revision, not a hypothetical example of team sprint.

3.4 Failure cases and source fidelity

Operating-profit paraphrase (T17). The question requires Group operating profit excluding significant items. BM25 cannot find the golden passage within five results. Dense ranks the page-1 table as the first result, and hybrid as the third one. The correct reference answer is reported EBIT of AUD 2,185 million, separate from AUD 2,754 million before significant items [1]. This is an observation of retrieval repair; no generated response was done.

Long financial passages. Without truncation, passage lengths in tokens are 647, 570, 531, 614, 604, 737, 1,054, 951, 1,400 and 202. Nine are over the limit of 512 tokens. A passage can be ranked high while the information that follows it was not captured in the embedding. Another table-aware chunking experiment is suggested. Row labels, year, units and footnote links should be retained, not just truncated by the word count.

Unsupported prompts. All eight unsupported questions obtain positive BM25 and TF-IDF scores, hence 0/8 is filtered out by the zero-overlap gate. Even if a particular year or organization is missing in the request, familiar terms like "revenue" and "Woolworths" can still match a historical document. The following statistic is not the hallucination rate for the LLM. It proves that positive similarity does not prove answerability.

Financial qualifications. Net debt is reported as AUD (15,989) million inclusive of leases, compared to AUD (4,115) million without leases but with derivatives. Total dividends are 84 cents/share or AUD 1,026 million; the total for 2024 includes a special dividend. Answers should preserve all the above qualifications. The seven comparisons involving numeric values were checked deterministically, such as AUD 469 million decrease in EBIT and AUD 485 million rise in the absolute value of total net-debt.

3.5 Results from the past and verification limits

The original audit of 18 questions by page proxy is still reproducible: BM25 Hit@5 is 17/18 and corrected NDCG@5 is 0.760. The similarity of rounded NDCG to the new benchmark is accidental because questions and labels differ.

The separate larger-corpus draft reports by the team the Hit@5 figures of 0.700, 0.900 and 0.800 for BM25, dense and hybrid, and also 70% answer correctness and 40% citation correctness [6]. These figures are not used to demonstrate the new findings. Eight-question and two-unanswerable question description of the paper contrasts with the latter all-answerable questions description of the paper, and there are no raw outputs presented. With eight answerable questions, the binary Hit@5 figure changes in increments of 12.5%; the reported increments need to be verified through the denominator.

The newly performed evaluation demonstrates comparative retrieval effectiveness, failed ablation, failed zero-overlap gate and a reproducible answer key. The accuracy of generated answers, agreement between humans, user interface behaviour and business time savings have not been evaluated.

4 Impact and significance of results

4.1 Value demonstrated and value still to validate

The verified contribution is a verifiable process by which to surface potential retrieval problems prior to relying on the answers that are generated for finance. This helps the decision-maker in prioritizing areas that require further engineering effort: rank the table and preserve its definition is a higher priority than improve answer fluency. Evidence panels in the interface provide an effective means of verification, even though its effectiveness with users has not yet been quantified.

Supervised analyst is a worthwhile investment of learning. Production deployment or claims of improvement to finance decisions have yet to be made. Dense addresses an important retrieval problem related to operating profit, but this alone does not demonstrate whether the evidence is being used correctly by the answer generator. The added problem of the unsupported question adds to the list of potential errors in that a retrieval match is possible without the desired answer.

4.2 Business-value measurement

A possible follow-up research may involve the recruitment of a limited number of investors or finance students as a pilot sample, provide details on their background experience, and reverse the sequence of performing the task manually or using the assistant. Matching yet different tasks should be used to avoid memorization, while both correctness of answers and time needed for verification should be measured, including hard questions without support. Individual paired comparisons and variance should be stated, not just an average.

The formula for annual net benefit in planning purposes can be given as $Q\Delta tC/60 - C_{run} - C_{maint}$,

where Q denotes annual number of tasks, Δt time savings per task verified in minutes and C hourly wage cost. There are no numbers plugged into the formula since none were gathered. Local hosting does not result in a cost-per-request hosted model fee but will have expenses related to hardware, energy, maintenance and verification.

4.3 Ethical, social and professional issues

Open financial data will reduce the amount of personally identifiable information in the dataset, however, there is still the possibility that user requests would contain some hints about their investments. Any operational application should limit the amount of requests stored and make users aware of the logging policy.

The most practical risk would be financial misinformation. The tool must be able to differentiate historical information from present information and keep the source dates and the accounting assumptions in the process. The tool must not make the lack of evidence become a recommendation. There would be an issue from the point of view of company disclosures because what is missing in one organization's reporting perspective does not mean that there is nothing happening.

It should also take into consideration that the users will vary in financial literacy and language proficiency. Checks that have been suggested are plain-language versus technical paraphrases, keyboard accessibility, readability of contrast and screen-reader labels. These are the evaluation obligations but not certification of accessibility and fairness. Professional practice should also require traceability of evidence, reproducibility of experiments, limitation discussion and model/data review before distribution. The notes of the source about the accounting mentioned AASB 8.

5 Project management and iterative process

5.1 Team organisation and reported development cycles

Milestone 1 specifies a Trello board and a GitHub repository and outlines a proposed plan for progressing through the process of data preparation, retrieval, generation, and evaluation [2]. In the draft version on 24 October, Vijaypal was assigned evaluation design and coordination, Raj was assigned retrieval and generation integration, and William was assigned data preparation, chunking, and result visualization [6]. These are functional assignments, but they do not reflect a percentage of contribution of each team member, which is documented separately.

This draft highlights four development cycles: the extraction of the bigger PDF into 494 chunks; implementation of keyword retrieval, dense retrieval, and hybrid retrieval; Llama 3.2 3B integration and prompting after fake rejections; and evaluation. It is a good outline of the expected cycles of iteration. However, there is no task export, commit log or meeting notes dated according to this outline, so it is impossible to confirm when these cycles took place or if acceptance tests proceeded implementation.

The most recent milestone feedback explicitly stated that access to public repositories and mentor invites to Trello and weekly reviews were required. However, those are operational concerns, and cannot be addressed through any prose revision. The final project log must reflect real-life interactions and feedback and associate tasks with commit messages and results. The GitHub address provided was evaluated for this revision. The archive on the main branch of the repository includes the retail customer churn data set and script but no RAG artifacts. The team should put up the RAG code and package on the proper repository; simply providing the link does not suffice as proof of submission.

5.2 Iteration driven by observed failures

Table 3: Evidence-based iteration record and next actions. Proposed actions are not claimed as completed.

Observation	Learning	Next action / acceptance evidence
Page equality is an incomplete relevance proxy	A page can contain both a table and unrelated notes	Independently adjudicate passage judgments and reference answers
Ideal DCG used only retrieved labels	Metric code can hide missing evidence	Audit evaluator uses corpus-wide ideal ranking; regression check supplied
Q06 misses required table	Term overlap is insufficient for reliable numerical QA	Try table-aware passages; rerun fixed tests and new holdout
Large passages mix tables and footnotes	Context quality may be lost before generation	Measure tokenizer lengths; keep units and linked notes when splitting
Source-count slider is disconnected	Visible controls can misrepresent actual settings	Pass selected value to pipeline; test observed context count
Embedding builder assumes output directory exists	Setup depends on execution order	Create directory explicitly and test a clean installation

The current iteration must link every problem to an acceptance criteria, make single changes and maintain the outcomes – whether they are failures or successes. The model, prompt, and questions need to be constant during retrieval experiments. The definition of done would involve reproducibility, execution of answers in an independent fashion and validation from the repository instead of just completing a demonstration. The updated backlog is: validate the new labels independently; experiment with table-aware chunking; execute and rate the generated answers; then find the user value. The undocumented and ambiguous prompts are now considered, and the retrieval comparison is completed.

5.3 Methodological reflection

What we learn from the audit is that the quality of the evaluation may be as limiting for progress as the quality of the models. High page hits hide low ranking information, duplication of the evidence, and ambiguities in the accounting. Better retrieval will not fix a wrongly labeled test set on its own. Likewise, a smaller size corpus makes it possible to check it but biases the retrieval since five hits account for half of the existing passages. The future work will have to refine the benchmark before expanding it, and keep the failed experiments to know why a modification worked or not.

6 Conclusion

The conducted diagnostic evaluation offers a firmer foundation for retrieval selection, where dense retrieval beats BM25 and equal-weighted hybrid rankings on the fixed corpus and 28 answerable questions. BM25 ablation proves the value of length normalization while the unsupported questions prove that zero overlap gating does not work here. These are all reproducible results with verifiable raw outputs and references.

The proposed pilot is dense retrieval with source verification but keeping uncertainties around unsupported and ambiguous queries open. Next gate release is answer validation and benchmark label checking independently. Business effectiveness or any reliable financial advice have to wait until the metrics from users are provided.

References

[1] Woolworths Group Limited. (2025). *2025 Five Year Summary*. Nine-page source PDF supplied with the project as `F25_Five_Year_Summary.pdf`. Financial tables and notes, pp. 1–9.

[2] Group 59. (2026). *WIL Project — Milestone 1 Report: Woolworths FY25 Report Q&A (RAG)*. Supplied course project report, two pages.

[3] Group 59 project materials. (Supplied 2026). *Woolworths RAG User Friendly Website*. Submitted ZIP containing Streamlit app, retrieval/generation code, source PDF, chunks and evaluation questions. The implementation inspected for this report; authorship allocation requires team confirmation.

[4] RMIT Information Retrieval Group. *Walert: Code and data for the Walert large language model-based chatbot*. `https://github.com/rmit-ir/walert`. Accessed 23 October 2026.

[5] Spina, D. *Walert — Your Open Day FAQ Buddy*. Project description. `https://www.damianospina.com/project/walert/`. Accessed 23 October 2026.

[6] Group 59. (24 October 2026). *WIL Project: Written Report*. Team draft supplied as `vijayWA+24_10(1).zip`. Larger-corpus configuration, reported metrics and development narrative; referenced image files and raw evaluation logs were not included.

[7] Sentence Transformers. *multi-qa-MiniLM-L6-cos-v1: Model card*. `https://huggingface.co/sentence-transformers/multi-qa-MiniLM-L6-cos-v1`. Accessed 23 October 2026.

A Repository, reproducibility and audit trail

Project artifacts: `https://github.com/s4097104/Data-Science-Assignment`. This is the latest repository address supplied by the team; public visibility, assessor access and upload of supporting artifacts require confirmation.

Project board: https://trello.com/invite/b/6a87f2ebd6838cb542ff80fd/ATTI11c8efd904ec4d752b3fdf17684d8841
project-group-59

B Question-level evidence on the new benchmark

Table 4: First relevant rank within five on the 28 answerable questions. Full question text and reference answers are in benchmark.csv; a dash denotes no hit.

ID	Type	BM25	Dense	Hybrid
T01	Direct	1	1	1
T02	Direct	2	1	1
T03	Direct	2	1	1
T04	Direct	1	1	1
T05	Direct	2	1	1
T06	Direct	5	1	2
T07	Direct	1	1	1

ID	Type	BM25	Dense	Hybrid
T08	Direct	1	1	1
T09	Direct	1	1	1
T10	Direct	1	1	1
T11	Direct	1	1	1
T12	Direct	1	1	1
T13	Paraphrase	1	1	1
T14	Paraphrase	4	1	2
T15	Paraphrase	4	1	2
T16	Paraphrase	1	1	1
T17	Paraphrase	–	1	3
T18	Paraphrase	2	1	1
T19	Paraphrase	1	1	1
T20	Paraphrase	5	2	1
T21	Comparison	4	3	–
T22	Comparison	1	1	1
T23	Comparison	5	2	3
T24	Comparison	1	1	1
T25	Comparison	1	1	1
T26	Comparison	1	1	1
T27	Comparison	1	3	2
T28	Comparison	1	1	1

The four ambiguous questions (T29–T32) and eight unsupported questions (T33–T40) are deliberately excluded from this table's answerable retrieval population. Their expected behaviours are provided in the benchmark, but no model responses have been scored.

C Reference-answer and regression examples

These source-checked examples retain the original audit's Q identifiers for traceability. The new benchmark uses T identifiers and supplies its complete reference-answer set separately. These examples are not model responses.

Table 5: Selected financial answer checks, based on the source PDF.

ID	Required content and interpretation
Q01	Group revenue is $69,077 million for 2025 (p. 1); distinguish revenue from profit.
Q06	Reported Group EBIT is $2,185 million; EBIT before significant items is $2,754 million (p. 1). State the basis or clarify ambiguity.
Q07	Profit attributable to equity holders after tax is $963 million (p. 2), not $1,385 million before significant items.
Q08–09	Total dividend is 84 cents per share / $1,026 million; final dividend is 45 cents per share (p. 6). Do not confuse declared dividends with cash-flow dividends paid.
Q11	$2,754m versus $3,223m; decrease $469m, approximately 14.6%. Disclose 52 versus 53 reporting weeks (p. 1).

ID	Required content and interpretation
Q13	Net cash provided by operating activities is $4,550 million (p. 5).
Q14	Total net debt is shown as $(15,989) million including leases; excluding lease liabilities and including derivatives it is $(4,115) million (p. 4).
Q15	Return on equity is 27.0% on the disclosed before-significant-items basis (pp. 6–7).
Q17–18	Basic EPS before significant items is 113.5 cents; Group ROFE is 13.7% using the percentage definition in note 3 (p. 6).

Further tests should ask for live prices, 2026 results and topics missing from this corpus; expected behaviour is to state insufficient evidence. Ambiguous profit/debt requests should trigger explicit definitions or clarification. Financial rounding should be accepted only when equivalent units, year and accounting basis are retained.

D Completed evaluation and outstanding validation

Table 6: Evidence register for the revised evaluation.

Item	Status
40-question benchmark	Completed with answerability labels, topic groups, reference answers and relevant passages; AI-assisted, independent human review pending.
Keyword/dense/hybrid comparison	Executed on the same 28 answerable questions; raw rankings, metric tables and model/runtime hashes included.
Length-normalisation ablation	Executed; removing normalisation reduced NDCG@5. Original setting retained.
Unsupported-query gate	Executed; zero-overlap rule rejected none of eight unsupported prompts. Not an LLM refusal score.
Numerical reference checks	Seven calculations verified in code; no generated responses scored.
LLM answer evaluation	Runner and blank scoring sheet supplied. Requires the team's Ollama environment and human scoring.
Historical larger experiment	Raw outputs, labels and screenshots still required before its reported figures can be confirmed.
Teamwork and business value	Personal reflections, mentoring/commit evidence and a user study require real team participation.